

Discrete-Time System Analysis Using the z-Transform

Chapter 7

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Outline

- The z-Transform
- The inverse z-Transform
- Properties of the z-Transform
- Z-Transform Solution of Linear Difference Equations
- Block Diagram and System Realization
- Frequency Response of Discrete-Time Systems
- System Design by Poles and Zeros Placement.

The Bilateral z-Transform

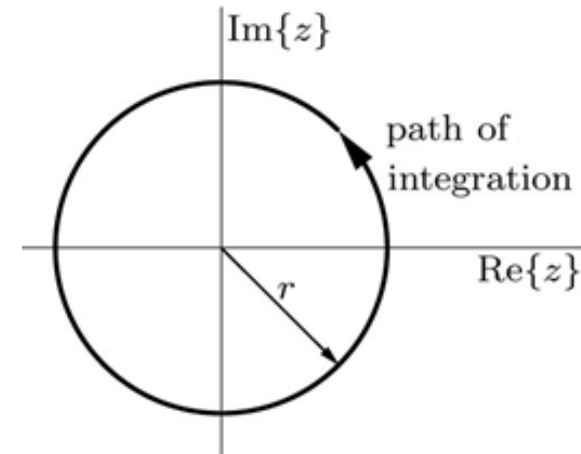
The z-transform is a mathematical tool in system analysis and design. It represent input $x[n]$ as a sum of everlasting exponentials (complex frequency) of the form z^n .

The Bilateral z-transform: $x[n]$ exist for positive and negative n .

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad z = re^{j\Omega}$$

The inverse z-transform

$$x[n] = \frac{1}{2\pi j} \oint X(z)z^{n-1}dz$$



$$X(z) = \mathcal{Z} \{x[n]\} \quad \text{and} \quad x[n] = \mathcal{Z}^{-1} \{X(z)\} \quad x[n] \xleftrightarrow{\mathcal{Z}} X(z)$$

Example

Find the z-transform and the corresponding ROC for the causal signal $x[n] = \gamma^n u[n]$.

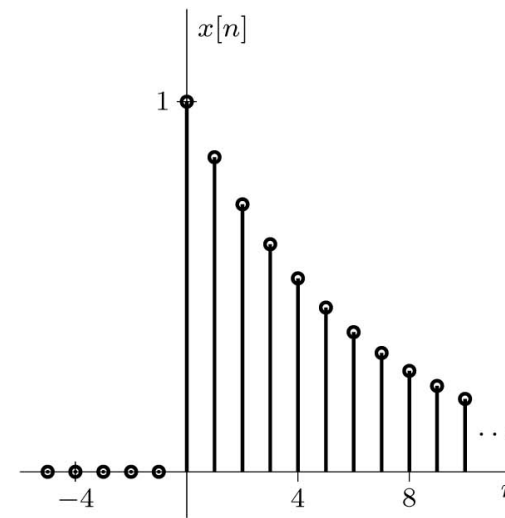
Solution

$$X(z) = \sum_{n=0}^{\infty} \left(\frac{\gamma}{z}\right)^n$$

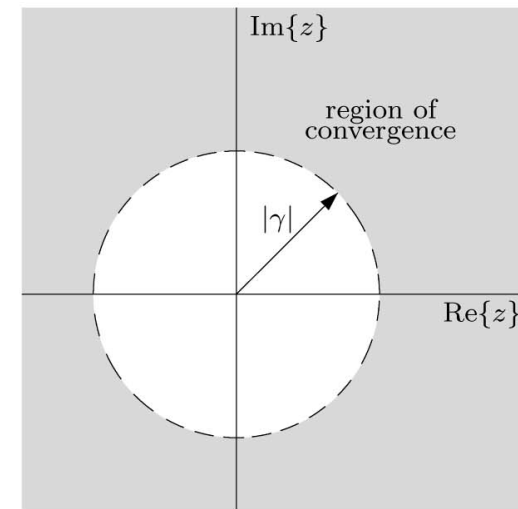
$$X(z) = \frac{1}{1 - \frac{\gamma}{z}} = \frac{z}{z - \gamma},$$

$$\text{ROC: } \left|\frac{\gamma}{z}\right| < 1 \quad \text{or} \quad |z| > |\gamma|$$

$$\sum_{m=p}^n r^m = \frac{r^p - r^{n+1}}{1-r}$$



(a)

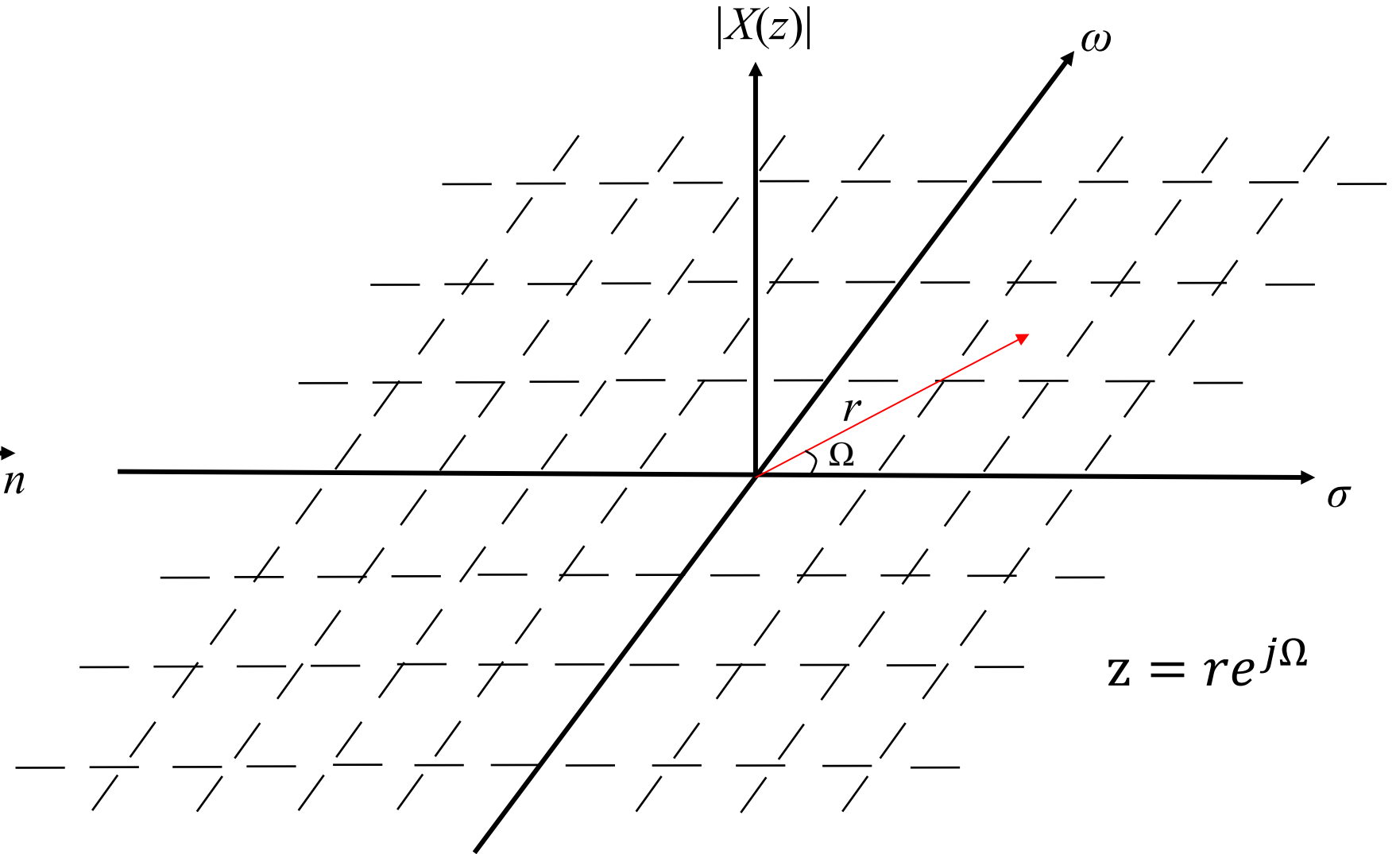
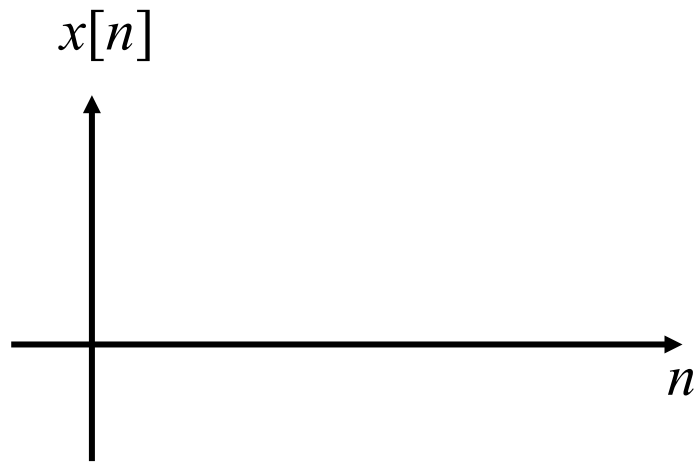


(b)

Note: for causal signal, the ROC extend outward to the centered circle.

Z – Domain and z-plane

Discrete Time Domain



Example

Find the z-transform and the corresponding ROC for the causal signal $y[n] = -\gamma^n u[-n-1]$.

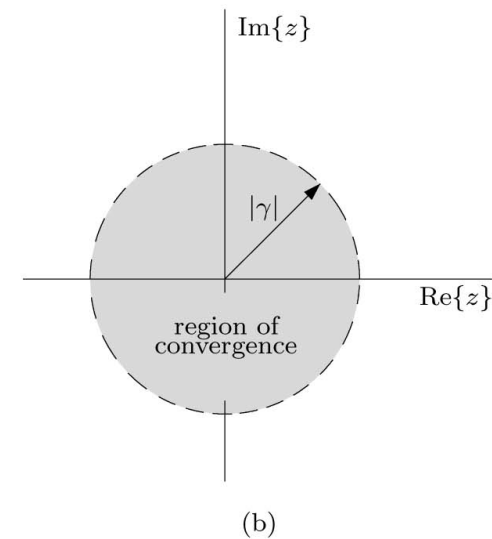
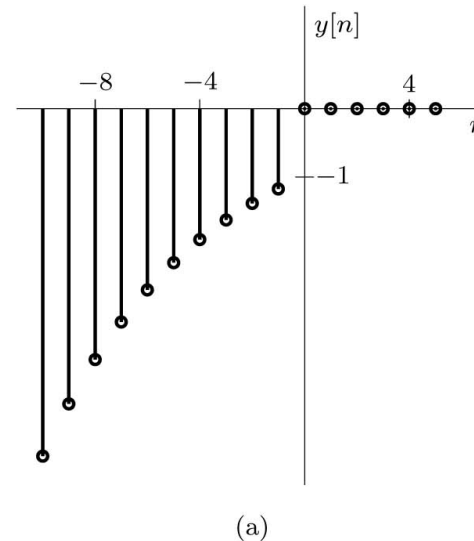
Solution

$$Y(z) = \sum_{n=-\infty}^{-1} -\left(\frac{\gamma}{z}\right)^n$$

$$Y(z) = \frac{z}{z - \gamma}$$

$$\text{ROC: } \left| \frac{z}{\gamma} \right| < 1 \quad \text{or} \quad |z| < |\gamma|$$

$$\sum_{m=p}^n r^m = \frac{r^p - r^{n+1}}{1-r}$$



Note: for anti-causal signal, the ROC extend inward to the centered circle.

Existence of the Bilateral z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \sum_{n=-\infty}^{\infty} \frac{x[n]}{z^n}$$

$$|X(z)| \leq \sum_{n=-\infty}^{\infty} \frac{|x[n]|}{|z|^n} < \infty$$

- The z-transform exist for any signal $x[n]$ that grows no faster than an exponential signal r^n , for some real and positive r .
- For a finite length sequence, the ROC is the entire z-plane except zero if $x[n]$ is nonzero for some positive n , and ∞ if $x[n]$ is nonzero for some negative n .
- If $z = x + y$ then the region of convergence for z (R_z) is at least $R_x \cap R_y$.

Example

Find the z-transform of $w[n] = x[n] + y[n]$ where $x[n] = (0.9)^n u[n]$ and $y[n] = (1.2)^n u[-n-1]$.
What is the z-transform of $w[n]$ if $x[n]$ is changed to $(2)^n u[n]$.

Solution

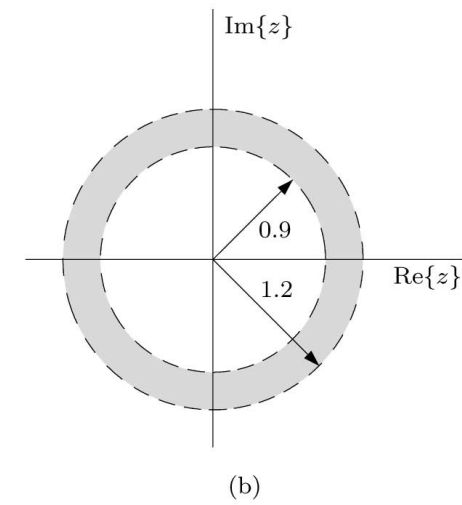
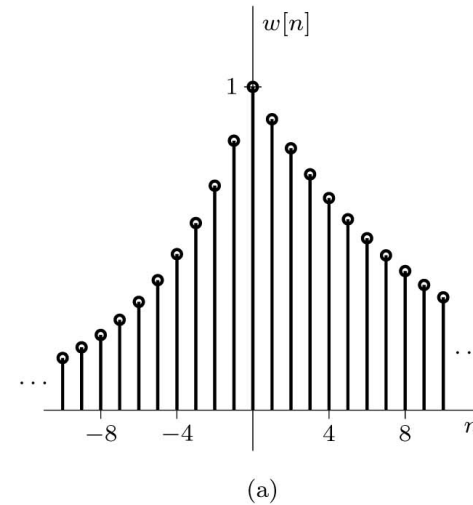
$$X(z) = \frac{z}{z - 0.9} \quad \text{with ROC } |z| > 0.9$$

$$Y(z) = \frac{-z}{z - 1.2} \quad \text{with ROC } |z| < 1.2$$

$$W(z) = X(z) + Y(z)$$

$$= \frac{z}{z - 0.9} + \frac{-z}{z - 1.2} = \frac{-0.3z}{(z - 0.9)(z - 1.2)}$$

$$\text{ROC: } 0.9 < |z| < 1.2$$



If $x[n]$ is changed to $(2)^n u[n]$ then the ROC of $X(z)$ becomes $|z| > 2$ and there is no common ROC for X and Y so the z-transform for w does not exist.

The Unilateral z-Transform

When the signal is causal, $x[n] = 0$ for $n < 0$, then the z-transform is unilateral:

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

Example: Determine the unilateral z-transforms of **(a)** $x_a[n] = \delta[n]$, **(b)** $x_b[n] = u[n]$,
(c) $x_c[n] = \cos(\beta n)u[n]$, and **(d)** $x_d[n] = u[n] - u[n - 5]$.

$$X_a(z) = 1, \quad \text{ROC: all } z$$

$$X_b(z) = \frac{z}{z - 1}, \quad \text{ROC: } \left| \frac{1}{z} \right| < 1 \quad \text{or} \quad |z| > 1$$

$$X_c(z) = \frac{z[z - \cos(\beta)]}{z^2 - 2\cos(\beta)z + 1}, \quad \text{ROC: } |z| > 1$$

$$X_d(z) = \frac{z}{z - 1}(1 - z^{-5}), \quad \text{ROC: all } z \neq 0$$

The transfer function $H(z)$ is the z-transform of the impulse response $h[n]$ of an LTID system;

$$h[n] \xLeftrightarrow{Z} H(z) \text{ with ROC } R_h$$

	$x[n]$	$X(z)$	ROC
1.	$\delta[n]$	1	All z
2.	$u[n]$	$\frac{z}{z-1}$	$ z > 1$
3.	$\gamma^n u[n]$	$\frac{z}{z-\gamma}$	$ z > \gamma $
4.	$\gamma^{n-1} u[n-1]$	$\frac{1}{z-\gamma}$	$ z > \gamma $
5.	$n\gamma^n u[n]$	$\frac{\gamma z}{(z-\gamma)^2}$	$ z > \gamma $
6.	$n^2 \gamma^n u[n]$	$\frac{\gamma z(z+\gamma)}{(z-\gamma)^3}$	$ z > \gamma $
7.	$\frac{n!}{(n-m)!m!} \gamma^{n-m} u[n]$	$\frac{z}{(z-\gamma)^{m+1}}$	$ z > \gamma $
8.	$ \gamma ^n \cos(\beta n) u[n]$	$\frac{z[z- \gamma \cos(\beta)]}{z^2-2 \gamma \cos(\beta)z+ \gamma ^2}$	$ z > \gamma $
9.	$ \gamma ^n \sin(\beta n) u[n]$	$\frac{z \gamma \sin(\beta)}{z^2-2 \gamma \cos(\beta)z+ \gamma ^2}$	$ z > \gamma $
10.	$ \gamma ^n \cos(\beta n + \theta) u[n]$	$\frac{z[z\cos(\theta)- \gamma \cos(\beta-\theta)]}{z^2-2 \gamma \cos(\beta)z+ \gamma ^2}$ $= \frac{(0.5e^{j\theta})z}{z- \gamma e^{j\beta}} + \frac{(0.5e^{-j\theta})z}{z- \gamma e^{-j\beta}}$	$ z > \gamma $
11.	$r \gamma ^n \cos(\beta n + \theta) u[n]$ $r = \sqrt{\frac{a^2 \gamma ^2+b^2-2abc}{ \gamma ^2-c^2}}$ $\beta = \cos^{-1}\left(\frac{-c}{ \gamma }\right)$ $\theta = \tan^{-1}\left(\frac{ac-b}{a\sqrt{ \gamma ^2-c^2}}\right)$	$\frac{z(az+b)}{z^2+2cz+ \gamma ^2}$	$ z > \gamma $
12.	$\delta[n-k]$	z^{-k}	$ z > 0 \quad k > 0$ $ z < \infty \quad k < 0$
13.	$-u[-n-1]$	$\frac{z}{z-1}$	$ z < 1$
14.	$-\gamma^n u[-n-1]$	$\frac{z}{z-\gamma}$	$ z < \gamma $
15.	$-n\gamma^n u[-n-1]$	$\frac{z\gamma}{(z-\gamma)^2}$	$ z < \gamma $

7.2

The Inverse z-Transform

Example

Using partial fraction expansions and the Table, determine the inverse bilateral z-transform of

$$X(z) = \frac{-z(z + 0.4)}{(z - 0.8)(z - 2)}$$

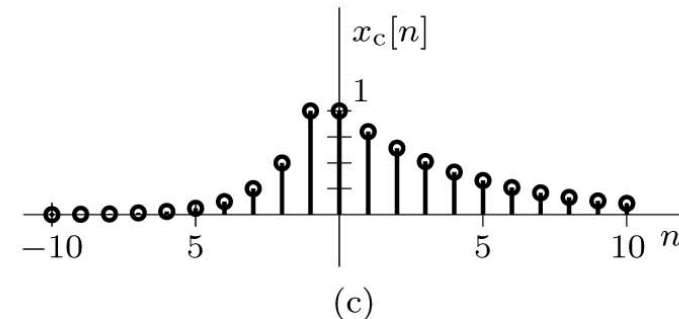
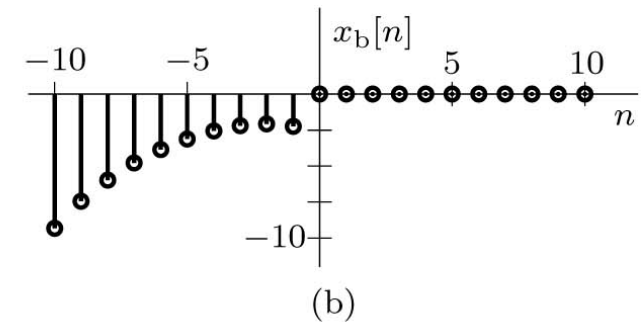
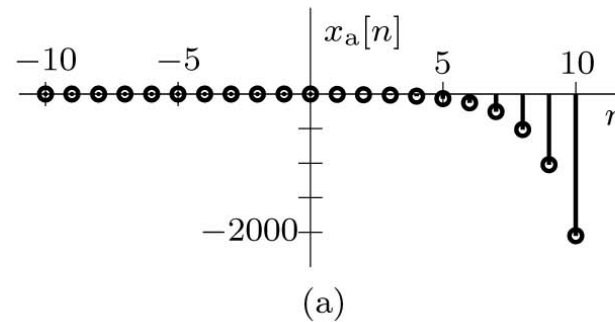
if the ROC is **(a)** $|z| > 2$, **(b)** $|z| < 0.8$, and **(c)** $0.8 < |z| < 2$.

Solution

$$x_a[n] = [(0.8)^n - 2(2)^n] u[n]$$

$$x_b[n] = [-(0.8)^n + 2(2)^n] u[-n - 1]$$

$$x_c[n] = (0.8)^n u[n] + 2(2)^n u[-n - 1]$$



The Inverse z-Transform

Using the contour integral to find the inverse z-transform require the knowledge of complex variable theory so we will use the table and partial fraction expansions to find the inverse z-transform.

▷ Example 7.5 (Inverse Unilateral z-Transform by Partial Fraction Expansion)

Using partial fraction expansions and Table 7.1, determine the inverse unilateral z-transforms of

$$(a) \quad X_a(z) = \frac{8z-19}{(z-2)(z-3)} \quad (b) \quad X_b(z) = \frac{z(2z^2-11z+12)}{(z-1)(z-2)^3} \quad (c) \quad X_c(z) = \frac{2z(3z+17)}{(z-1)(z^2-6z+25)}$$

$$x_a[n] = [3(2)^{n-1} + 5(3)^{n-1}] u[n-1] \quad \text{or} \quad x_a[n] = -\frac{19}{6}\delta[n] + \left[\frac{3}{2}(2)^n + \frac{5}{3}(3)^n\right] u[n]$$

$$x_b[n] = -\left[3 + \frac{1}{4}(n^2 + n - 12)2^n\right] u[n]$$

$$x_c[n] = [2 + 3.2016(5)^n \cos(0.9273n - 2.2455)] u[n]$$

Read example
in Textbook

7.3

Properties of the z-Transform

Properties of z-Transform

Linear Property

$$ax[n] + by[n] \xLeftrightarrow{\mathcal{Z}_u} aX(z) + bY(z)$$

Time Scaling Property

$$\gamma^n u[n] \longleftrightarrow \frac{z}{z-\gamma}$$

$$\text{if } x[n] \xLeftrightarrow{\mathcal{Z}} X(z), \text{ then } x_{\uparrow}[n] \xLeftrightarrow{\mathcal{Z}} X(z^L)$$

Time-Reversal Property

$$\text{if } x[n] \xLeftrightarrow{\mathcal{Z}} X(z) \text{ with ROC } R_x, \text{ then } x[-n] \xLeftrightarrow{\mathcal{Z}} X(1/z) \text{ with ROC } 1/R_x$$

Properties of z-Transform

Bilateral z-Transform Time-Shifting Property

if $x[n] \xleftrightarrow{\mathcal{Z}} X(z)$ with ROC R_x , then $x[n - m] \xleftrightarrow{\mathcal{Z}} z^{-m} X(z)$ with ROC almost R_x

Unilateral z-Transform Time-Shifting Property

Right Shift (Delay)

$$x[n - m]u[n - m] \xleftrightarrow{\mathcal{Z}_u} z^{-m} X(z), \quad (m > 0)$$

$$x[n - m]u[n] \xleftrightarrow{\mathcal{Z}_u} z^{-m} X(z) + z^{-m} \sum_{n=1}^m x[-n]z^n, \quad (m > 0)$$

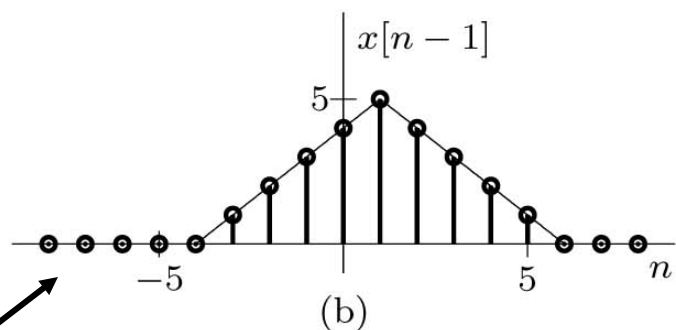
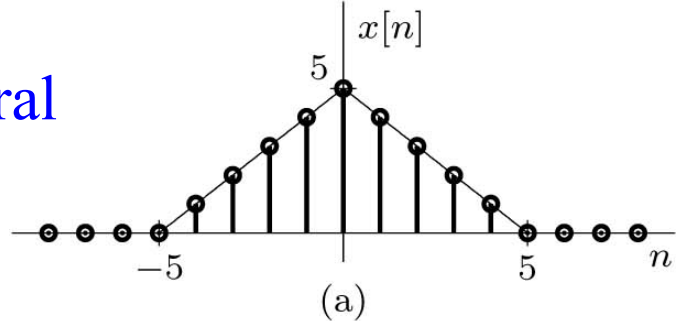
$$x[n - 1]u[n] \xleftrightarrow{\mathcal{Z}_u} z^{-1} X(z) + x[-1].$$

$$x[n - 2]u[n] \xleftrightarrow{\mathcal{Z}_u} z^{-2} X(z) + z^{-1}x[-1] + x[-2]$$

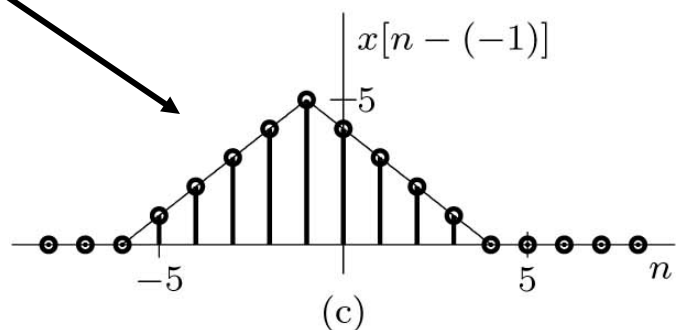
Left Shift (Advance)

$$x[n + m]u[n] \xleftrightarrow{\mathcal{Z}_u} z^m X(z) - z^m \sum_{n=0}^{m-1} x[n]z^{-n} \quad (m > 0)$$

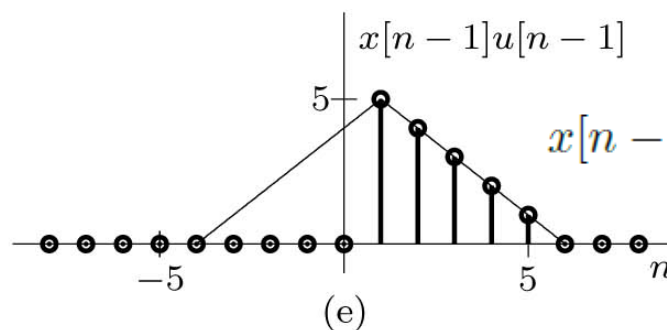
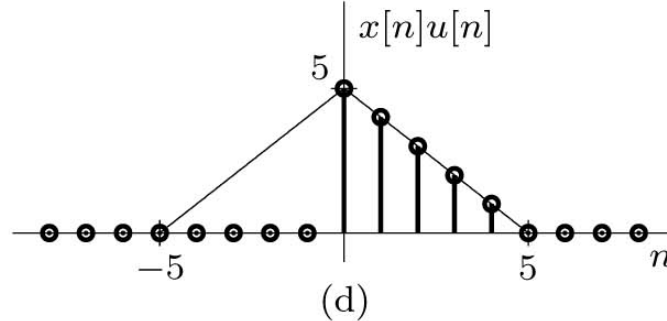
Bilateral



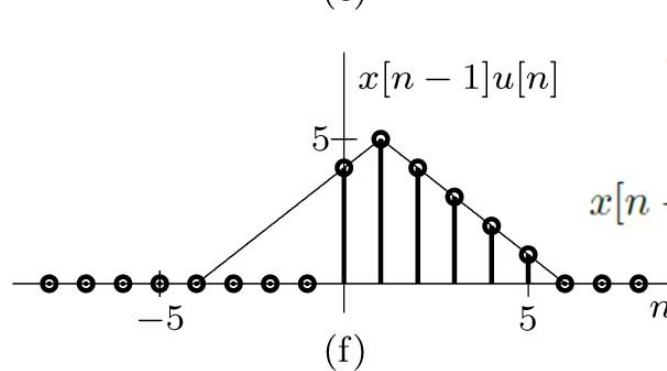
if $x[n] \xLeftrightarrow{\mathcal{Z}} X(z)$ with ROC R_x ,
 then $x[n-m] \xLeftrightarrow{\mathcal{Z}} z^{-m}X(z)$ with ROC almost R_x



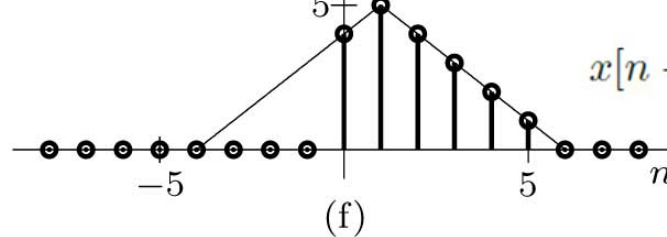
Unilateral



$$x[n-m]u[n-m] \xLeftrightarrow{\mathcal{Z}_u} z^{-m}X(z), \quad (m > 0)$$



$$x[n-1]u[n] \xLeftrightarrow{\mathcal{Z}_u} z^{-1}X(z) + x[-1] \quad (m > 0)$$



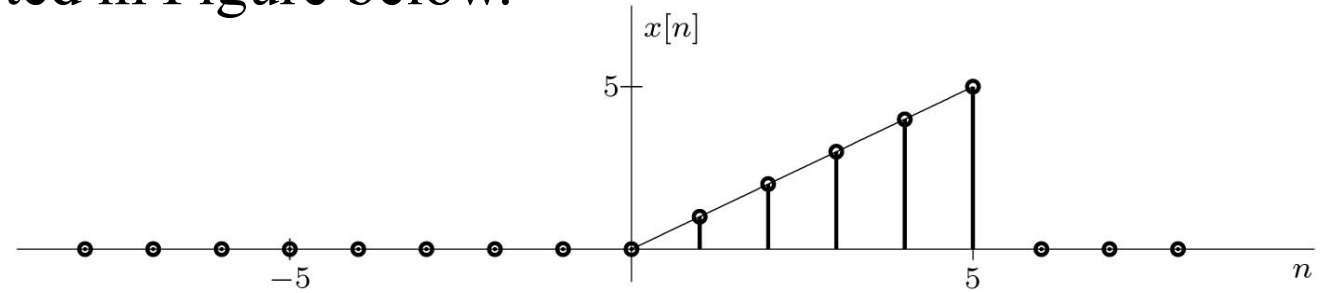
$$x[n-m]u[n] \xLeftrightarrow{\mathcal{Z}_u} z^{-m}X(z) + z^{-m} \sum_{n=1}^m x[-n]z^n$$

$$x[n+m]u[n] \xLeftrightarrow{\mathcal{Z}_u} z^mX(z) - z^m \sum_{n=0}^{m-1} x[n]z^{-n} \quad (m > 0)$$

Example

Find the z-transform of the signal $x[n]$ depicted in Figure below.

Solution



$$x[n] = nu[n] - \{(n - 6)u[n - 6] + 6u[n - 6]\}$$

$$X(z) = \frac{z}{(z - 1)^2} - \frac{1}{z^5(z - 1)^2} - \frac{6}{z^5(z - 1)}$$

$$X(z) = \frac{z^6 - 6z + 5}{z^5(z - 1)^2}$$

$$n\gamma^n u[n] \xLeftrightarrow{\mathcal{Z}} \frac{\gamma z}{(z - \gamma)^2}$$

Properties of z-Transform

Z-Domain Scaling Property

If $x[n] \xleftrightarrow{\mathcal{Z}} X(z)$ with ROC R_x ,
then $\gamma^n x[n] \xleftrightarrow{\mathcal{Z}} X\left(\frac{z}{\gamma}\right)$ with ROC $|\gamma|R_x$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Z-Domain Differentiation Property

If $x[n] \xleftrightarrow{\mathcal{Z}} X(z)$ with ROC R_x ,
then $nx[n] \xleftrightarrow{\mathcal{Z}} -z \frac{d}{dz} X(z)$ with ROC R_x

$u[n]$	$\longleftrightarrow$	$\frac{z}{z-1}$	$ z > 1$
$\gamma^n u[n]$	$\longleftrightarrow$	$\frac{z}{z-\gamma}$	$ z > \gamma $

Z-Domain Convolution Property

If $x[n] \xleftrightarrow{\mathcal{Z}} X(z)$ with ROC R_x and $y[n] \xleftrightarrow{\mathcal{Z}} Y(z)$ with ROC R_y ,
then $x[n] * y[n] \xleftrightarrow{\mathcal{Z}} X(z)Y(z)$ with ROC at least $R_x \cap R_y$

Bilateral z -Transform	Unilateral z -Transform
Synthesis: $x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$ Analysis: $X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}, \text{ ROC: } R_x$ Linearity: $ax[n] + by[n] \xLeftrightarrow{\mathcal{Z}} aX(z) + bY(z),$ ROC: At least $R_x \cap R_y$ Complex Conjugation: $x^*[n] \xLeftrightarrow{\mathcal{Z}} X^*(z^*), \text{ ROC: } R_x$ Time Reversal: $x[-n] \xLeftrightarrow{\mathcal{Z}} X(1/z), \text{ ROC: } 1/R_x$ Time Shifting: $x[n-m] \xLeftrightarrow{\mathcal{Z}} z^{-m}X(z), \text{ ROC: Almost } R_x$ z-Domain Scaling: $\gamma^n x[n] \xLeftrightarrow{\mathcal{Z}} X(z/\gamma), \text{ ROC: } \gamma R_x$ z-Domain Differentiation: $nx[n] \xLeftrightarrow{\mathcal{Z}} -z \frac{d}{dz} X(z), \text{ ROC: } R_x$ Time Convolution: $x[n] * y[n] \xLeftrightarrow{\mathcal{Z}} X(z)Y(z), \text{ ROC: At least } R_x \cap R_y$	Synthesis: $x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$ Analysis: $X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$ Linearity: $ax[n] + by[n] \xLeftrightarrow{\mathcal{Z}_u} aX(z) + bY(z)$ Complex Conjugation: $x^*[n] \xLeftrightarrow{\mathcal{Z}_u} X^*(z^*)$ Time Reversal: Time Shifting: If $m > 0$: $x[n-m]u[n-m] \xLeftrightarrow{\mathcal{Z}_u} z^{-m}X(z)$ (general case given below) z-Domain Scaling: $\gamma^n x[n] \xLeftrightarrow{\mathcal{Z}_u} X(z/\gamma)$ z-Domain Differentiation: $nx[n] \xLeftrightarrow{\mathcal{Z}_u} -z \frac{d}{dz} X(z)$ Time Convolution: $x[n] * y[n] \xLeftrightarrow{\mathcal{Z}_u} X(z)Y(z)$
Unilateral z-Transform Time Shifting, General Case If $m > 0$: $x[n-m]u[n] \xLeftrightarrow{\mathcal{Z}_u} z^{-m}X(z) + z^{-m} \sum_{n=1}^m x[-n]z^n$ If $m < 0$: $x[n-m]u[n] \xLeftrightarrow{\mathcal{Z}_u} z^{-m}X(z) - z^{-m} \sum_{n=0}^{-m-1} x[n]z^{-n}$	

7.4

z-Transform Solution of Linear Difference Equations

Example: Total Response

When initial conditions are present, the unilateral z-transform is generally the appropriate analysis tool. When only the zero-state response is required, either the bilateral or unilateral z-transform may be appropriate.

Example: Given input $x[n] = (0.5)^n u[n]$ and initial conditions $y[-1] = 11/6$ and $y[-2] = 37/36$, use the unilateral z-transform to solve the second-order ($K = 2$) constant-coefficient linear difference equation (find $y[n]$).

$$y[n + 2] - 5y[n + 1] + 6y[n] = 3x[n + 1] + 5x[n].$$

Solution

$$\frac{Y(z)}{z} = \frac{(3z^2 - 9.5z + 10.5)}{(z - 0.5)(z^2 - 5z + 6)}$$

$$Y(z) = \frac{26}{15} \left(\frac{z}{z - 0.5} \right) - \frac{7}{3} \left(\frac{z}{z - 2} \right) + \frac{18}{5} \left(\frac{z}{z - 3} \right)$$

$$y[n] = \left[\frac{26}{15} (0.5)^n - \frac{7}{3} (2)^n + \frac{18}{5} (3)^n \right] u[n]$$

$$x[n - m]u[n] \xLeftrightarrow{\mathcal{Z}_u} z^{-m} X(z) + z^{-m} \sum_{n=1}^m x[-n]z^n \quad (m > 0)$$

Example: Zero-Input and Zero-State Components

Example: Given input $x[n] = (0.5)^n u[n]$ and initial conditions $y[-1] = 11/6$ and $y[-2] = 37/36$, use the unilateral z-transform to solve the second-order ($K = 2$) constant-coefficient linear difference equation

$$y[n + 2] - 5y[n + 1] + 6y[n] = 3x[n + 1] + 5x[n].$$

Solution

$$(1 - 5z^{-1} + 6z^{-2}) Y(z) - \underbrace{(3 - 11z^{-1})}_{\text{IC terms}} = \underbrace{\frac{3}{z - 0.5} + \frac{5}{z(z - 0.5)}}_{\text{input terms}}$$

$$(z^2 - 5z + 6) Y(z) = \underbrace{z(3z - 11)}_{\text{IC terms}} + \underbrace{\frac{z(3z + 5)}{(z - 0.5)}}_{\text{input terms}} \quad Y(z) = \underbrace{\frac{z(3z - 11)}{z^2 - 5z + 6}}_{\text{ZIR}} + \underbrace{\frac{z(3z + 5)}{(z - 0.5)(z^2 - 5z + 6)}}_{\text{ZSR}}$$

$$y[n] = \underbrace{[5(2)^n - 2(3)^n] u[n]}_{\text{ZIR}} + \underbrace{\left[\frac{26}{15}(0.5)^n - \frac{22}{3}(2)^n + \frac{28}{5}(3)^n \right] u[n]}_{\text{ZSR}}$$

The Transfer Function and ZSR of LTID

The Difference
Equation of a System

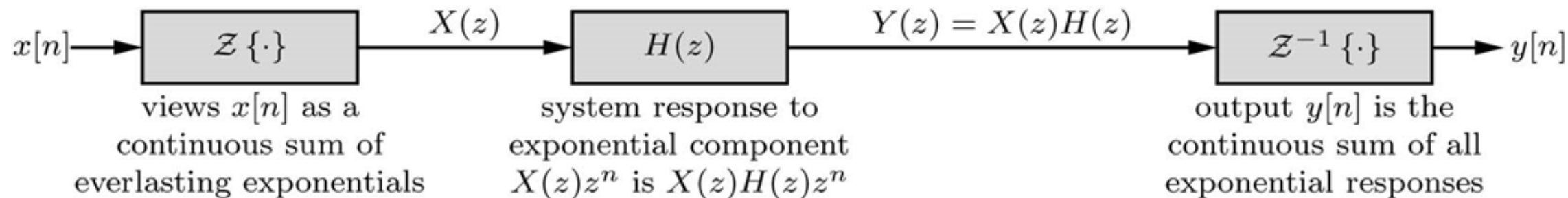
$$y[n + K] + a_1 y[n + (K - 1)] + \cdots + a_{K-1} y[n + 1] + a_K y[n] = b_0 x[n + K] + b_1 x[n + (K - 1)] + \cdots + b_{K-1} x[n + 1] + b_K x[n]$$

Take the z-transform, shifting property, and setting all IC to zero the above equation will be

$$(z^K + a_1 z^{K-1} + \cdots + a_{K-1} z + a_K) Y(z) = (b_0 z^K + b_1 z^{K-1} + \cdots + b_{K-1} z + b_K) X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{B(z)}{A(z)} = \frac{b_0 z^K + b_1 z^{K-1} + \cdots + b_{K-1} z + b_K}{z^K + a_1 z^{K-1} + \cdots + a_{K-1} z + a_K}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\mathcal{Z} \{ \text{zero-state response} \}}{\mathcal{Z} \{ \text{input} \}}$$



Example for Zero-State Response

Example: Given input $x[n] = (-2)^{-n}u[n]$, use the z -transform to determine the zero-state response $y[n]$ of a causal LTID system described by the difference equation

$$y[n + 2] + y[n + 1] + 0.16y[n] = x[n + 1] + 0.32x[n].$$

Solution

ROC ?

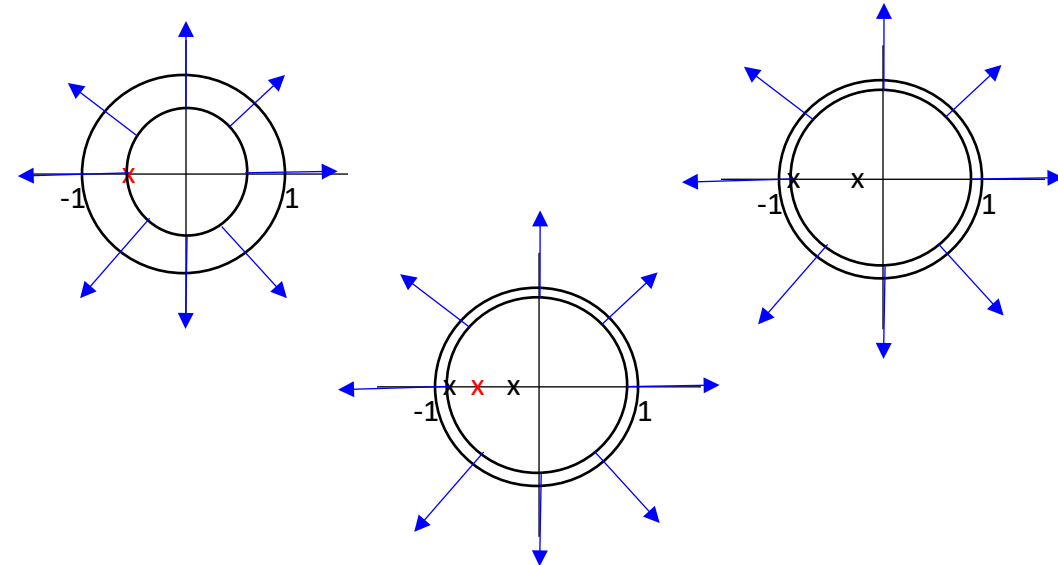
$$H(z) = \frac{B(z)}{A(z)} = \frac{z + 0.32}{z^2 + z + 0.16}$$

$$X(z) = \frac{z}{z + 0.5}$$

$$Y(z) = X(z)H(z) = \frac{z(z + 0.32)}{(z^2 + z + 0.16)(z + 0.5)}$$

$$Y(z) = \frac{2}{3} \left(\frac{z}{z + 0.2} \right) - \frac{8}{3} \left(\frac{z}{z + 0.8} \right) + 2 \left(\frac{z}{z + 0.5} \right)$$

$$y[n] = \left[\frac{2}{3}(-0.2)^n - \frac{8}{3}(-0.8)^n + 2(-0.5)^n \right] u[n]$$



Example with Causal and Non-causal Inputs

Example 3: Given two-sided input $x[n] = (0.8)^n u[n] + 2(2)^n u[-n - 1]$, use the z-transform to determine the zero-state response $y[n]$ of a causal LTID system described by the transfer function $H(z) = z / (z - 0.5)$.

Solution

$$\mathcal{Z} \{ (0.8)^n u[n] \} = \frac{z}{z - 0.8} \quad \text{with ROC } |z| > 0.8 \qquad \mathcal{Z} \{ 2(2)^n u[-n - 1] \} = -\frac{2z}{z - 2} \quad \text{with ROC } |z| < 2$$

$$X(z) = \frac{-z(z + 0.4)}{(z - 0.8)(z - 2)}, \quad \text{ROC: } 0.8 < |z| < 2$$

$$Y(z) = X(z)H(z) = \frac{-z^2(z + 0.4)}{(z - 0.5)(z - 0.8)(z - 2)}, \quad \text{ROC: } 0.8 < |z| < 2$$

$$Y(z) = -\left(\frac{z}{z - 0.5} \right) + \frac{8}{3} \left(\frac{z}{z - 0.8} \right) - \frac{8}{3} \left(\frac{z}{z - 2} \right) \quad \text{with ROC } 0.8 < |z| < 2$$

$$y[n] = \left[-(0.5)^n + \frac{8}{3}(0.8)^n \right] u[n] + \frac{8}{3}(2)^n u[-n - 1]$$

[Read Example](#)

Example with Inputs with Disjoint ROC

Example 4: For the system $H(z) = z / (z - 0.5)$, find the zero-state response to the input

$$x[n] = \underbrace{(0.8)^n u[n]}_{x_1[n]} + \underbrace{(0.6)^n u[-n-1]}_{x_2[n]}.$$

Solution

$$X_1(z) = \frac{z}{z - 0.8} \quad \text{with ROC } |z| > 0.8$$

$$X_2(z) = -\frac{z}{z - 0.6} \quad \text{with ROC } |z| < 0.6$$

There is no common ROC between $x_1[n]$ and $x_2[n]$ so use superposition to find the system response to both inputs separately.

$$Y_1(z) = \frac{z^2}{(z - 0.5)(z - 0.8)} = -\frac{5}{3} \left(\frac{z}{z - 0.5} \right) + \frac{8}{3} \left(\frac{z}{z - 0.8} \right) \quad \text{with ROC } |z| > 0.8$$

$$Y_2(z) = \frac{-z^2}{(z - 0.5)(z - 0.6)} = 5 \left(\frac{z}{z - 0.5} \right) - 6 \left(\frac{z}{z - 0.6} \right) \quad \text{with ROC } 0.5 < |z| < 0.6$$

$$y_1[n] = \left[-\frac{5}{3}(0.5)^n + \frac{8}{3}(0.8)^n \right] u[n]$$

$$y_2[n] = 5(0.5)^n u[n] + 6(0.6)^n u[-n-1]$$

Read Example

System Stability and the Transfer Function $H(z)$

- A causal LTID system is asymptotically stable if and only if all the characteristic roots are inside the unit circle. The roots may be simple or repeated.
- A causal LTID system is marginally stable if and only if there are no roots outside the unit circle and there are non-repeated roots on the unit circle.
- A causal LTID system is unstable if and only if at least one root is outside the unit circle, there are repeated roots on the unit circle, or both.

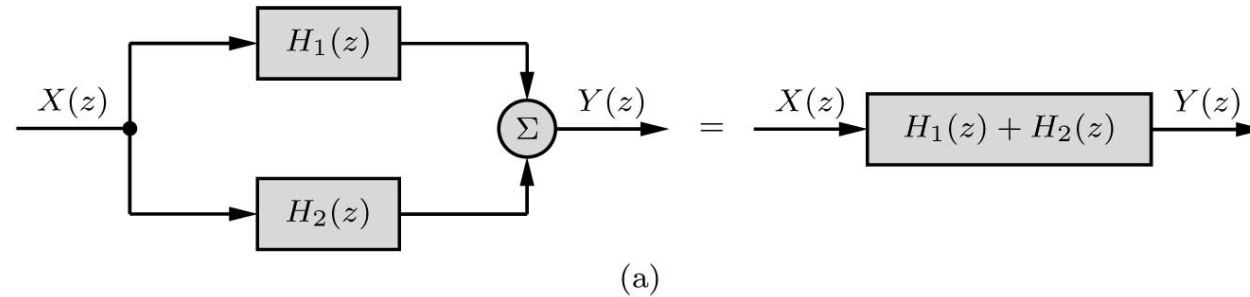
7.5

Block Diagrams and System Realization

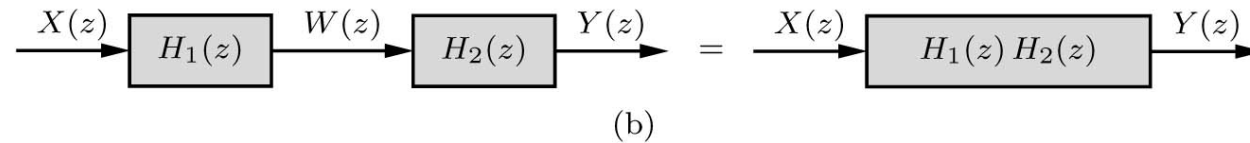
Basic Connections

The following is true if there is no loading effect between connected subsystems.

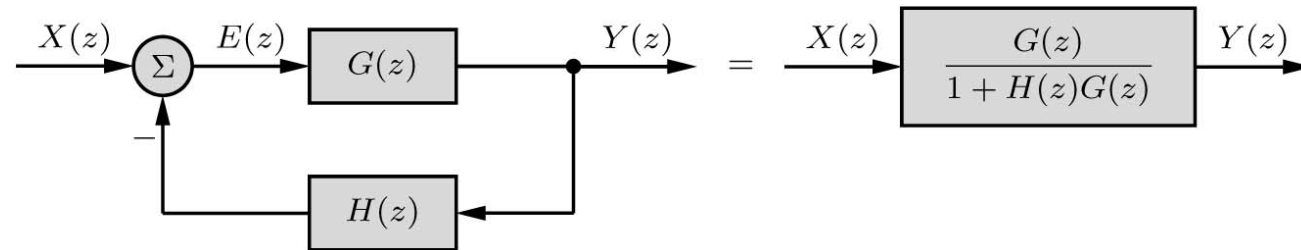
Parallel Connection



Cascade Connection



Feedback System



Direct Form Realization

$$H(z) = \frac{b_0 z^L + b_1 z^{L-1} + \dots + b_{L-1} z + b_L}{z^K + a_1 z^{K-1} + \dots + a_{K-1} z + a_K}$$

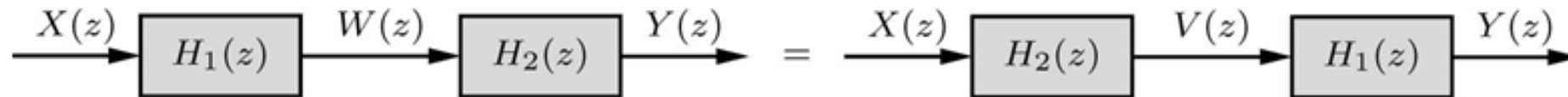
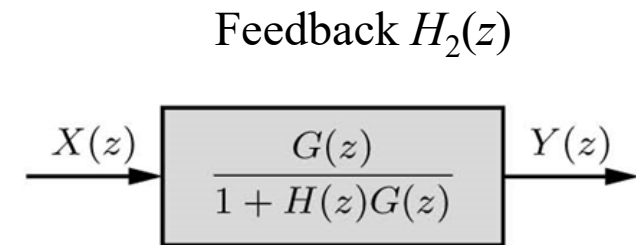
For causal systems, $L \leq K$. Multiply numerator and denominator by z^{-K} . Next express $H(z)$ as a cascade of two systems.

Example:

$$H(z) = \frac{b_0 z^2 + b_1 z + b_2}{z^2 + a_1 z + a_2}$$

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

$$H(z) = \underbrace{(b_0 + b_1 z^{-1} + b_2 z^{-2})}_{H_1(z)} \underbrace{\left(\frac{1}{1 + a_1 z^{-1} + a_2 z^{-2}} \right)}_{H_2(z)}$$



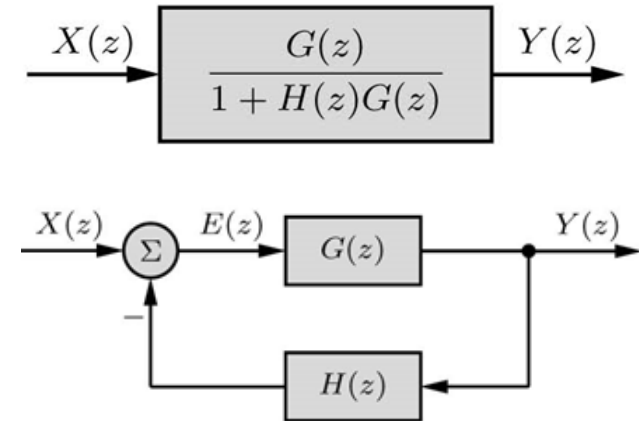
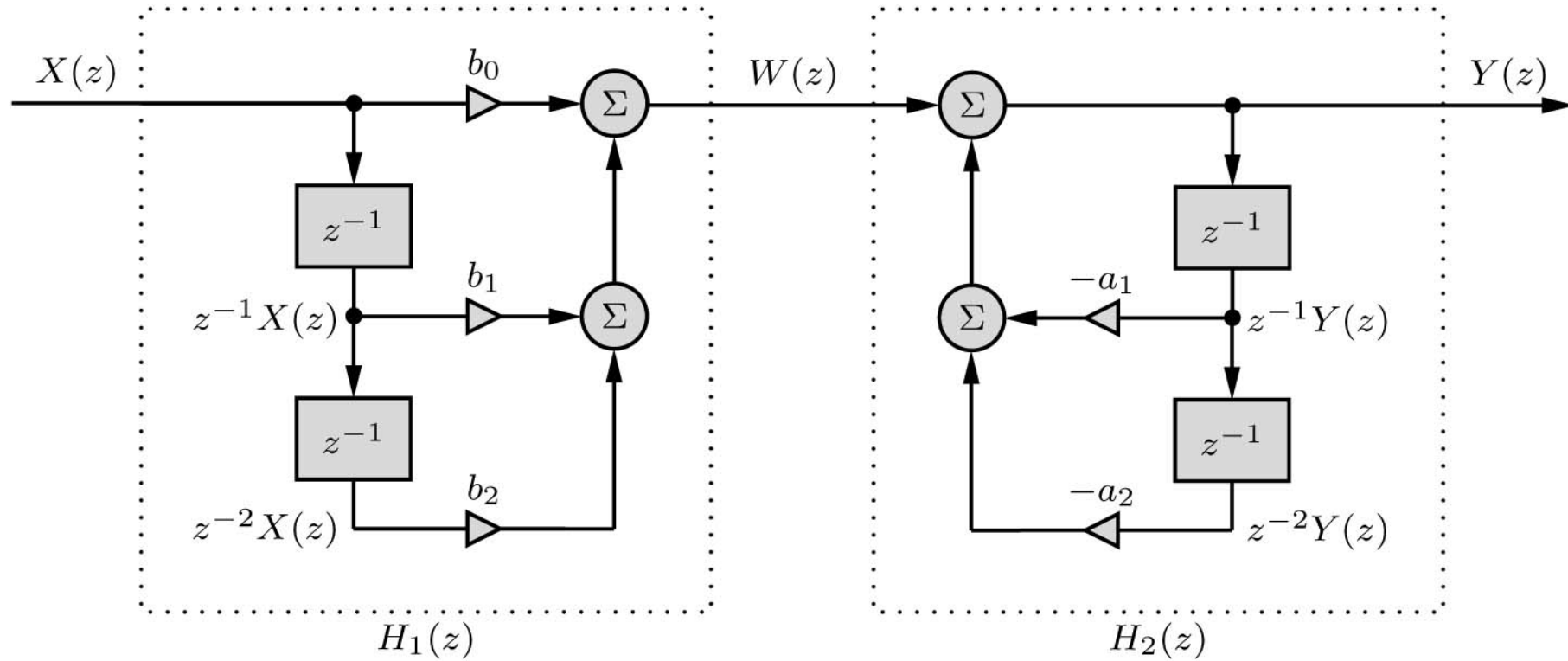
Direct Form I

Direct Form II

Direct Form I Realization

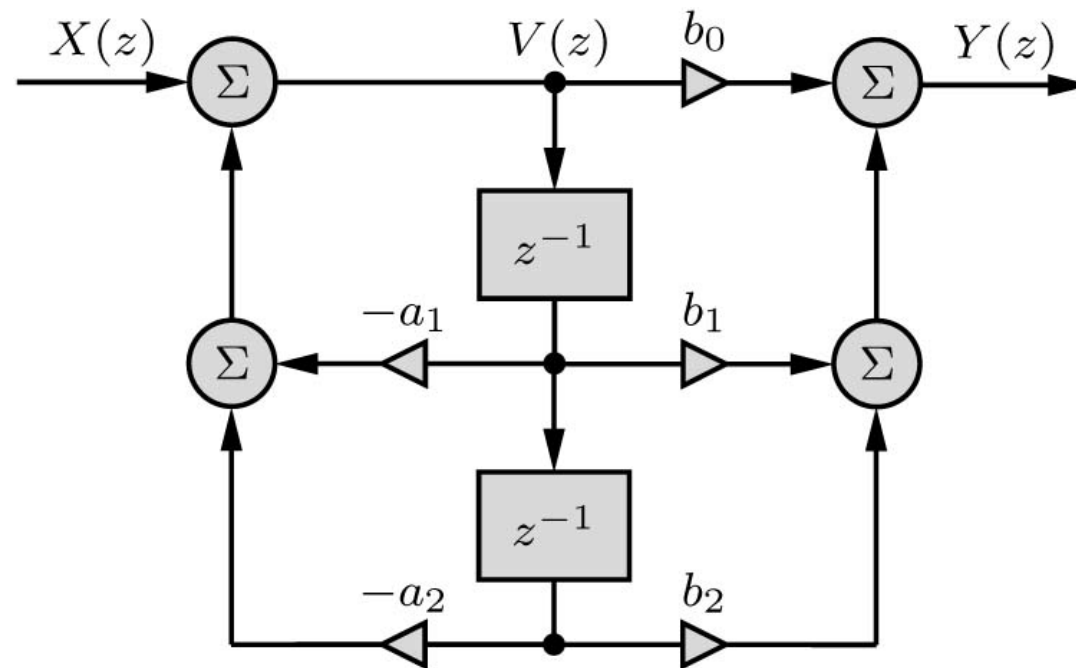
$$H(z) = \underbrace{(b_0 + b_1 z^{-1} + b_2 z^{-2})}_{H_1(z)} \underbrace{\left(\frac{1}{1 + a_1 z^{-1} + a_2 z^{-2}} \right)}_{H_2(z)}$$

$X(z) \rightarrow \boxed{H_1(z)} \xrightarrow{W(z)} \boxed{H_2(z)} \rightarrow Y(z)$



Direct Form II Realization

$$H(z) = \underbrace{\left(\frac{1}{1 + a_1 z^{-1} + a_2 z^{-2}} \right)}_{H_2(z)} \underbrace{(b_0 + b_1 z^{-1} + b_2 z^{-2})}_{H_1(z)} \xrightarrow{X(z)} \boxed{H_2(z)} \xrightarrow{V(z)} \boxed{H_1(z)} \xrightarrow{Y(z)}$$

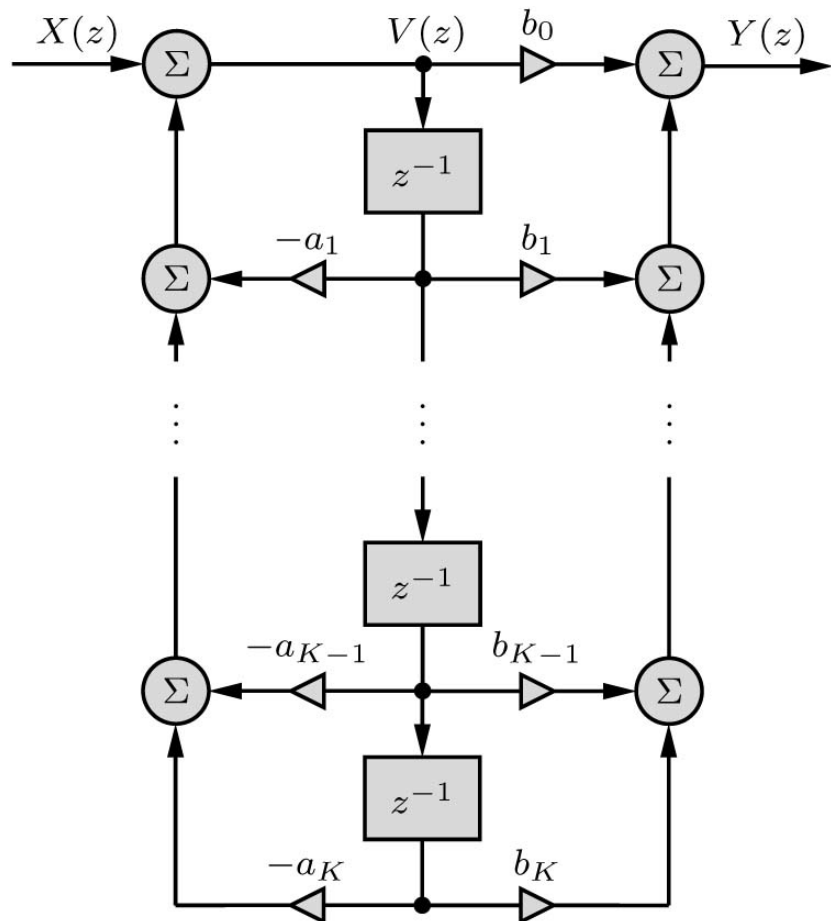


DFII is **canonic** since the number of delays equal to the order of the transfer function.

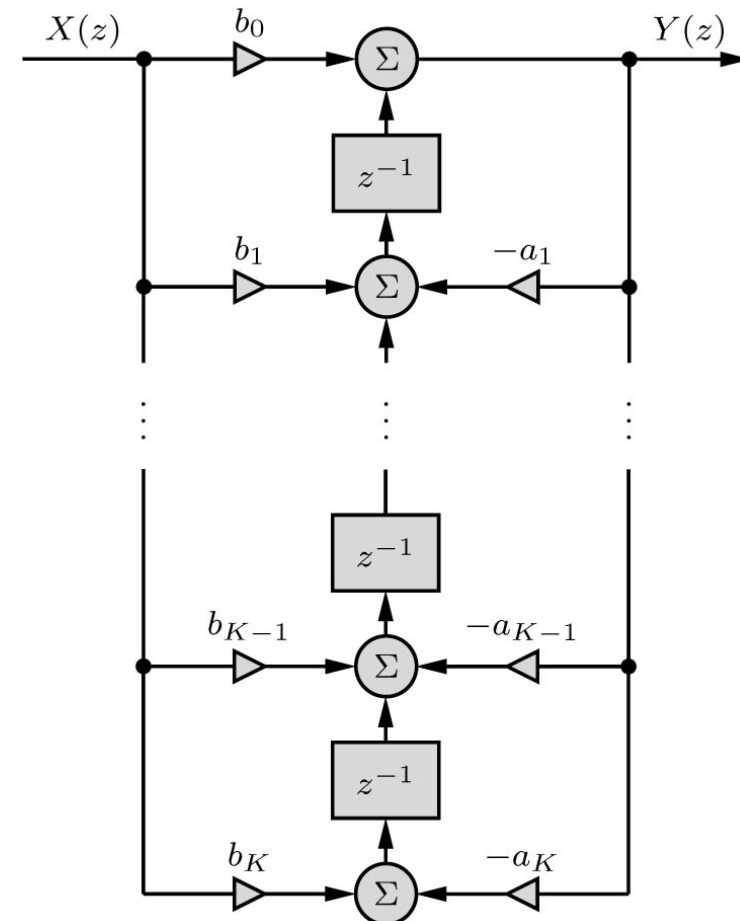
Transpose Realization (TDFII)

$$H(z) = \frac{b_0 z^L + b_1 z^{L-1} + \dots + b_{L-1} z + b_L}{z^K + a_1 z^{K-1} + \dots + a_{K-1} z + a_K}$$

Direct Form II (DFII)



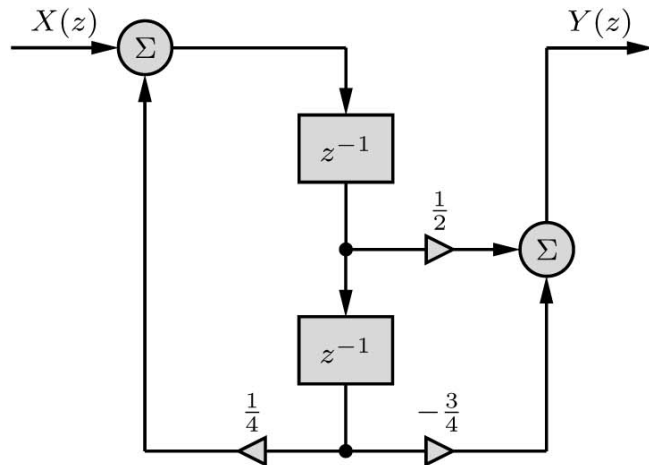
Transpose Direct Form II (TDFII)



Example

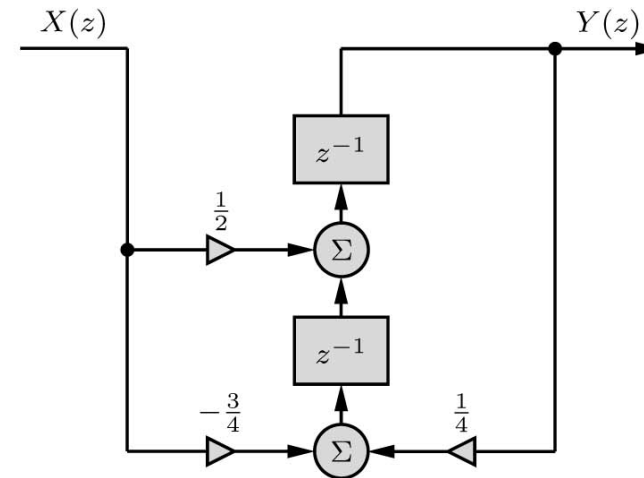
Find the DFII and TDFII realizations of an LTID system with transfer function

$$H(z) = \frac{2z - 3}{4z^2 - 1}$$



(a)

Direct Form II



(b)

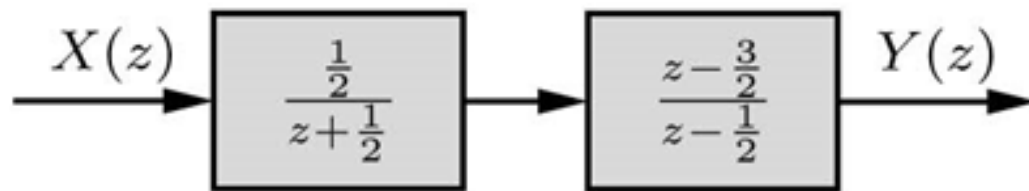
Transpose Direct Form II

Cascade and Parallel Realization

$$H(z) = \frac{2z - 3}{4z^2 - 1} = \frac{1/2(z - 3/2)}{(z + 1/2)(z - 1/2)}$$

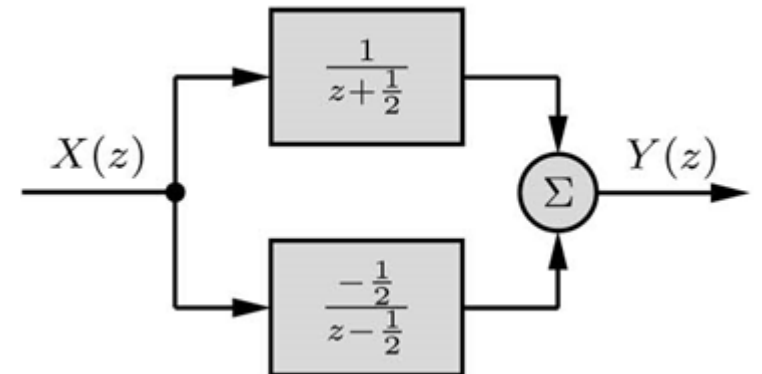
Cascade

$$H(z) = \underbrace{\left(\frac{1/2}{z + 1/2} \right)}_{H_1(z)} \underbrace{\left(\frac{z - 3/2}{z - 1/2} \right)}_{H_2(z)}$$



Parallel

$$H(z) = \underbrace{\frac{1}{z + 1/2}}_{H_3(z)} + \underbrace{\frac{-1/2}{z - 1/2}}_{H_4(z)}$$



Realization of Complex-Conjugate Roots

$$H(z) = \frac{z^3 + z}{16z^3 - 28z^2 + 20z - 6}$$

$$H(z) = \left(\frac{\frac{1}{16}z}{z - \frac{3}{4}} \right) \left(\frac{z + j}{z - (\frac{1}{2} + j\frac{1}{2})} \right) \left(\frac{z - j}{z - (\frac{1}{2} - j\frac{1}{2})} \right)$$

Cascade

$$H(z) = \left(\frac{\frac{1}{16}z}{z - \frac{3}{4}} \right) \left(\frac{z^2 + 1}{z^2 - z + \frac{1}{2}} \right)$$

Parallel

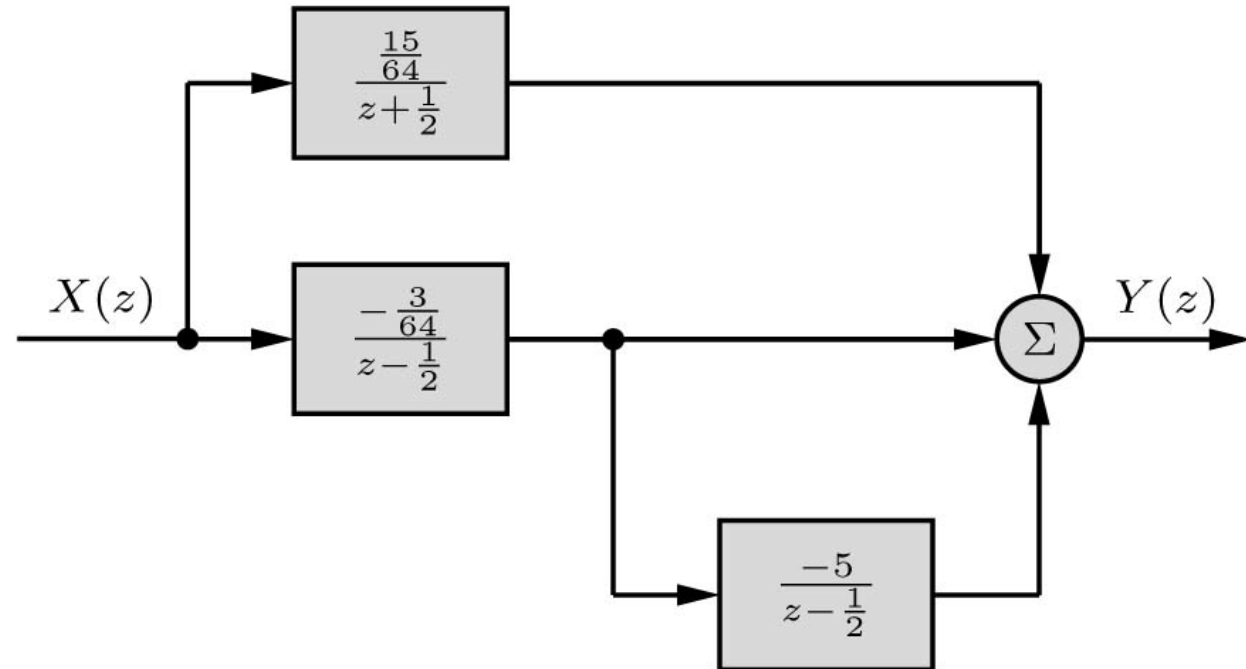
$$H(z) = \frac{\frac{5}{16}z}{z - \frac{3}{4}} + \frac{-\frac{1}{4}z^2 + \frac{1}{8}z}{z^2 - z + \frac{1}{2}}$$

Realization of Repeated Roots

For parallel connection you need to reuse some of the blocks so the number of delay equal the order of the transfer function.

Example: Determine a parallel realization of $H(z) = \frac{\frac{3}{16}(z^2 + 1)}{(z + 1/2)(z - 1/2)^2}$

$$H(z) = \frac{-\frac{3}{64}}{z - \frac{1}{2}} + \frac{\frac{15}{64}}{(z - \frac{1}{2})^2} + \frac{\frac{15}{64}}{z + \frac{1}{2}}$$



Realization of FIR Filters

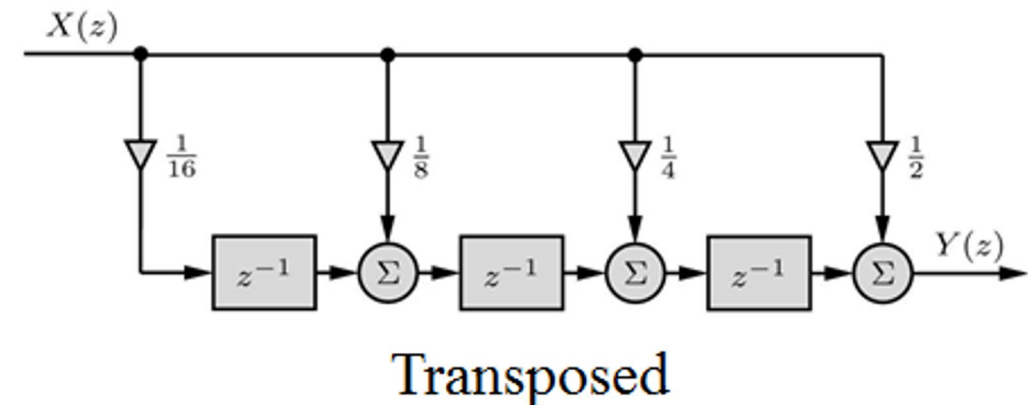
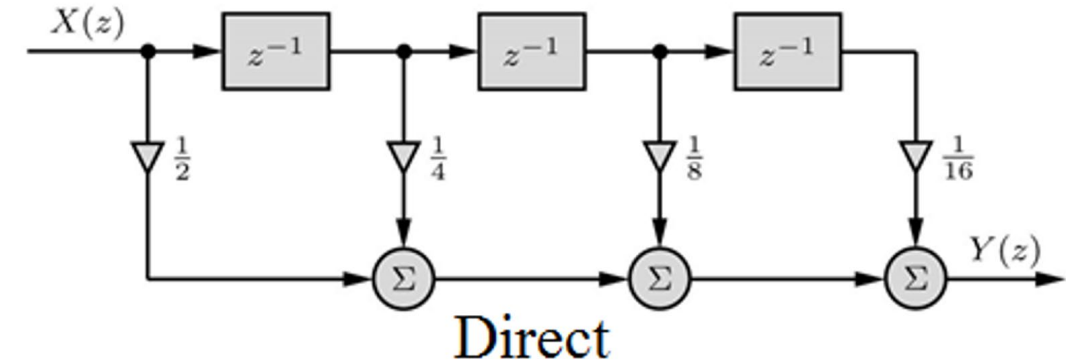
For FIR filters, a_0 is normalized to unity, and the remaining coefficients $a_k = 0$ for all $k \neq 0$.

$$y[n + K] + a_1 y[n + (K - 1)] + \cdots + a_{K-1} y[n + 1] + a_K y[n] = \\ b_0 x[n + K] + b_1 x[n + (K - 1)] + \cdots + b_{K-1} x[n + 1] + b_K x[n]$$

Example: Using direct and transposed direct forms to realize the FIR filter

$$y[n] = \frac{1}{2}x[n] + \frac{1}{4}x[n - 1] + \frac{1}{8}x[n - 2] + \frac{1}{16}x[n - 3]$$

$$H(z) = \frac{1}{2} + \frac{1}{4}z^{-1} + \frac{1}{8}z^{-2} + \frac{1}{16}z^{-3}$$



Do All Realization Lead to the Same Performance?

Theoretically all realization are equivalent if parameters are implemented with infinite precision.

- The finite word-length errors that plague these implementations include coefficient quantization, overflow errors, and round-off errors.
- From a practical viewpoint, parallel and cascade forms using low-order filters minimize finite word-length effects.
- In practice, high-order filters are most commonly realized using a cascade of multiple second-order sections, which are not only easier to design but are also less susceptible to coefficient quantization and round-off errors.

7.6

Frequency Response and Design of Discrete-Time Systems

System Response to Everlasting Sinusoid

The response of a real and asymptotically or BIBO stable LTID system to a sinusoidal input (or exponential) is the same sinusoid (exponential), modified only in gain and phase.

$$x[n] = z_o^n = (\gamma e^{j\Omega_o})^n$$

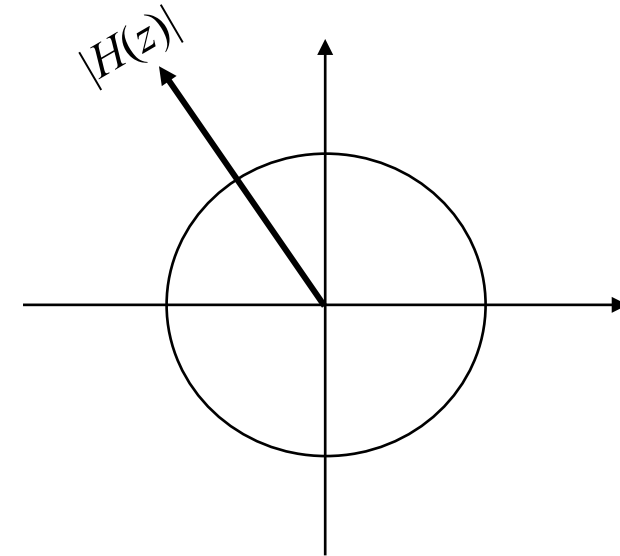
$$\begin{aligned} x[n] = z_o^n &\longrightarrow \boxed{H(z)} \longrightarrow \\ &= e^{j\Omega_o n} \\ &= \cos(\Omega_o n + \theta) \end{aligned}$$

$$y[n] = H(z_o)z_o^n$$

$$y[n] = H(e^{j\Omega_o})e^{j\Omega_o n}$$

$$y[n] = H(e^{j\Omega_o})\cos(\Omega_o n + \theta)$$

$$y[n] = |H(e^{j\Omega_o})|\cos(\Omega_o n + \theta + \angle H(e^{j\Omega_o}))$$



Steady-State Response to Causal Sinusoidal Inputs

$$y_{ss}[n] = |H(e^{j\Omega})|\cos[\Omega n + \theta + \angle H(e^{j\Omega})]u[n]$$

Example

Determine the frequency response $H(e^{j\Omega})$ of the system specified by the equation

$$y[n + 1] - 0.8y[n] = x[n + 1].$$

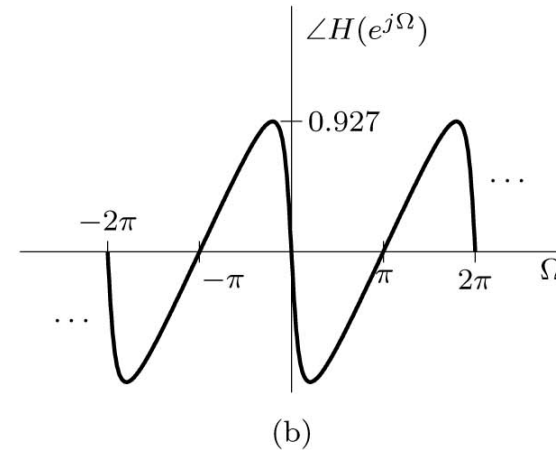
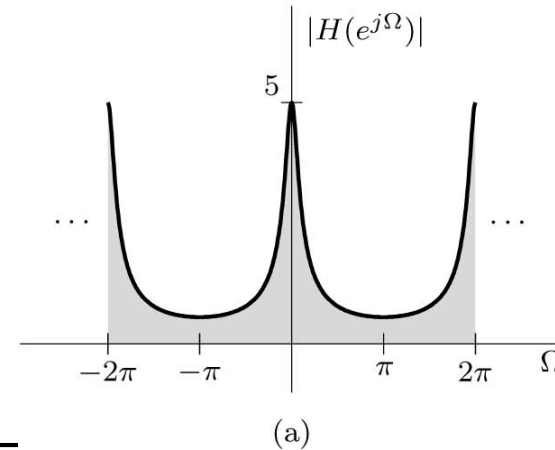
Determine the system responses to the inputs **(a)** $x_a[n] = 1^n = 1$, **(b)** $x_b[n] = \cos(\pi/6 n - 0.2)$, and **(c)** $x_c(t) = \cos(1500t)$ sampled using sampling interval $T = 0.001$.

Solution

$$H(z) = \frac{1}{1 - 0.8z^{-1}}$$

$$H(e^{j\Omega}) = \frac{1}{1 - 0.8e^{-j\Omega}} = \frac{1}{1 - 0.8\cos(\Omega) + j0.8\sin(\Omega)}$$

$$|H(e^{j\Omega})| = \frac{1}{\sqrt{1.64 - 1.6\cos(\Omega)}} \quad \text{and} \quad \angle H(e^{j\Omega}) = -\tan^{-1} \left(\frac{0.8\sin(\Omega)}{1 - 0.8\cos(\Omega)} \right)$$



Read Example in Textbook

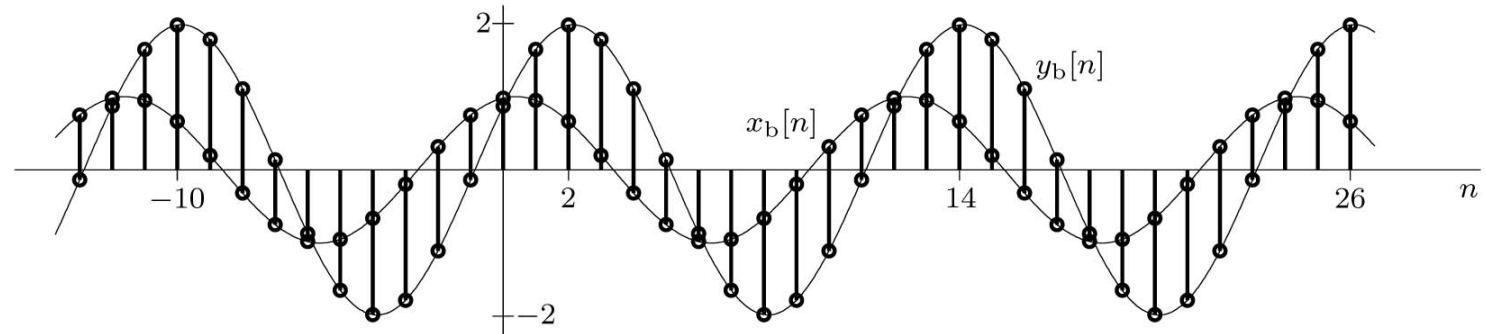
Continue Example

$$|H(e^{j\Omega})| = \frac{1}{\sqrt{1.64 - 1.6 \cos(\Omega)}} \quad \text{and} \quad \angle H(e^{j\Omega}) = -\tan^{-1} \left(\frac{0.8 \sin(\Omega)}{1 - 0.8 \cos(\Omega)} \right)$$

$$y_a[n] = 5$$

$$y_b[n] = 1.983 \cos(\pi/6 n - 1.116)$$

$$y_c[n] = 0.8093 \cos(1.5 n - 0.702)$$



```
Omega = linspace(-2*pi,2*pi,500); H = 1./(1-0.8*exp(-1j*Omega));  
subplot(121); plot(Omega,abs(H)); subplot(122); plot(Omega,angle(H));
```

Frequency Response from Pole-Zero Locations

$$H(z) = b_0 \frac{(z - z_1)(z - z_2) \cdots (z - z_K)}{(z - p_1)(z - p_2) \cdots (z - p_K)} = b_0 \frac{\prod_{l=1}^K (z - z_l)}{\prod_{k=1}^K (z - p_k)}$$

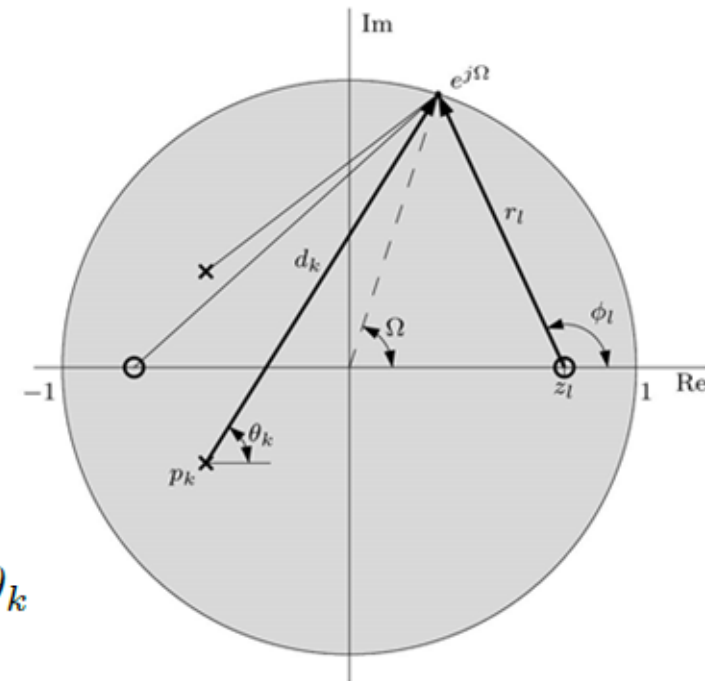
$$\begin{aligned} H(e^{j\Omega}) &= H(z)|_{z=e^{j\Omega}} = b_0 \frac{(r_1 e^{j\phi_1})(r_2 e^{j\phi_2}) \cdots (r_K e^{j\phi_K})}{(d_1 e^{j\theta_1})(d_2 e^{j\theta_2}) \cdots (d_K e^{j\theta_K})} = b_0 \frac{\prod_{l=1}^K r_l e^{j\phi_l}}{\prod_{k=1}^K d_k e^{j\theta_k}} \\ &= b_0 \frac{r_1 r_2 \cdots r_K}{d_1 d_2 \cdots d_K} e^{j[(\phi_1 + \phi_2 + \cdots + \phi_K) - (\theta_1 + \theta_2 + \cdots + \theta_K)]} \end{aligned}$$

$$\begin{aligned} |H(e^{j\Omega})| &= |b_0| \frac{r_1 r_2 \cdots r_K}{d_1 d_2 \cdots d_K} = |b_0| \frac{\prod_{l=1}^K r_l}{\prod_{k=1}^K d_k} \\ &= |b_0| \frac{\text{product of the distances of zeros to } e^{j\Omega}}{\text{product of the distances of poles to } e^{j\Omega}}, \end{aligned}$$

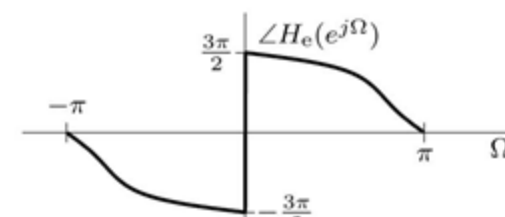
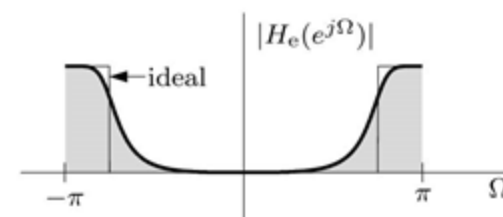
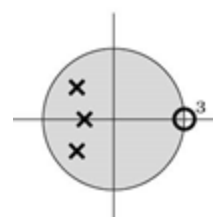
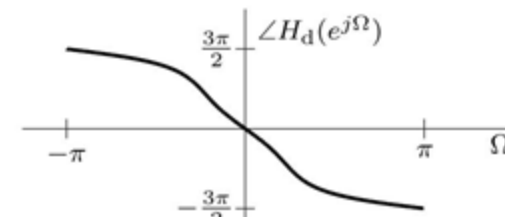
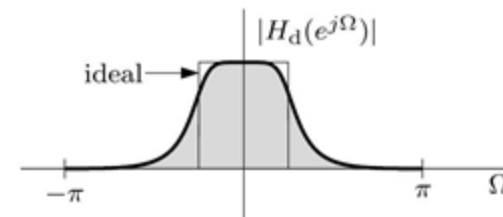
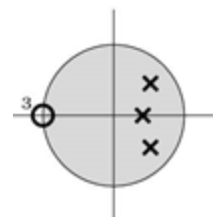
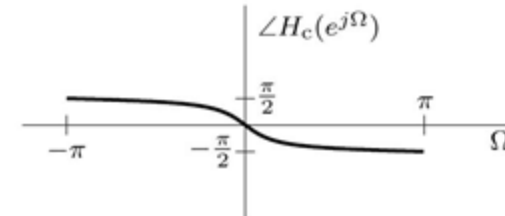
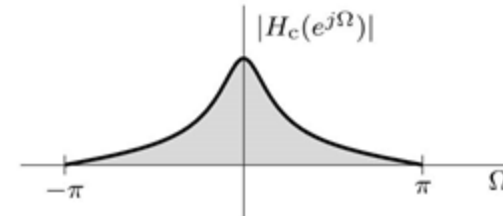
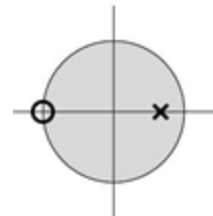
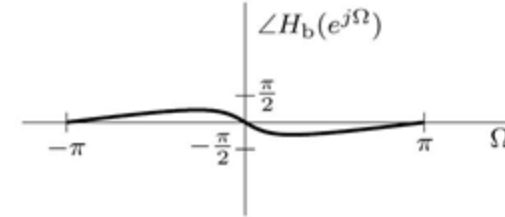
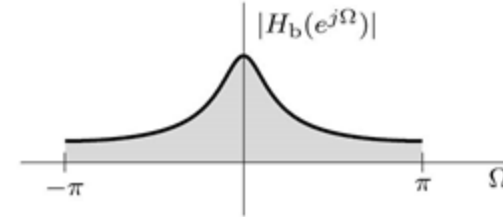
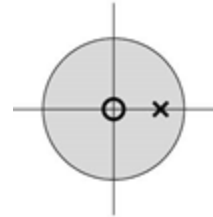
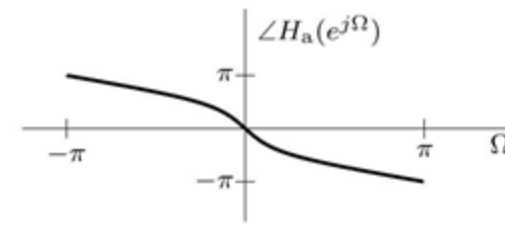
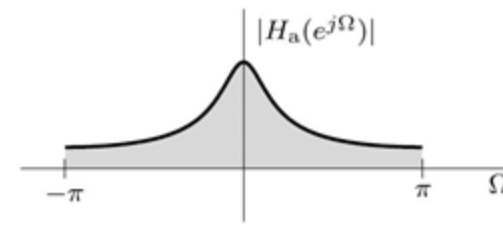
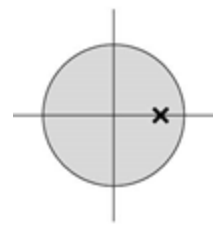
$$\begin{aligned} \angle H(e^{j\Omega}) &= \angle b_0 + (\phi_1 + \phi_2 + \cdots + \phi_K) - (\theta_1 + \theta_2 + \cdots + \theta_K) = \angle b_0 + \sum_{l=1}^K \phi_l - \sum_{k=1}^K \theta_k \\ &= \angle b_0 + \text{sum of zero angles to } e^{j\Omega} - \text{sum of pole angles to } e^{j\Omega} \end{aligned}$$

$$H(z) = \frac{z^2 - 0.81}{z^2 + 1.6z + 0.8}$$

$$H(z) = \frac{(z - 0.9)(z + 0.9)}{(z + 0.8 - j0.4)(z + 0.8 + j0.4)}$$



- To enhance the magnitude response at a frequency Ω , place a pole close to the point $e^{j\Omega}$.
- Placing a pole or a zero at the origin does not influence the magnitude response, but it adds angle $-\Omega$ (or Ω) to the phase response $\angle H(e^{j\Omega})$.
- To suppress the magnitude response at a frequency Ω , we should place a zero close to the point $e^{j\Omega}$.
- Repeating poles or zeros further enhances their influence.
- Placing a zero close to a pole tends to cancel the effect of that pole on the frequency response (and vice versa).
- For a stable system, all the poles must be located inside the unit circle.



Example

Design a tuned (bandpass) analog filter with zero transmission at 0 Hz and also at the highest frequency $f_{\max} = 500$ Hz. The resonant frequency is required to be 125 Hz.

Solution

$$f_s \geq 2f_{\max} = 1000 \text{ Hz}$$

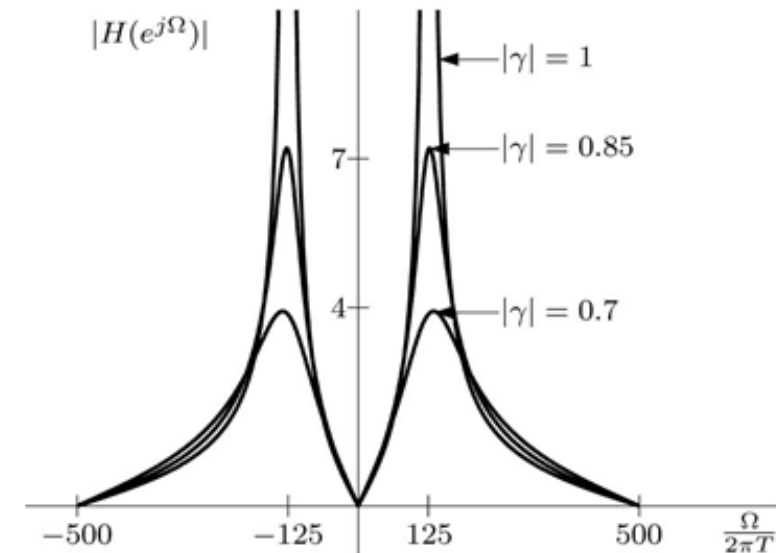
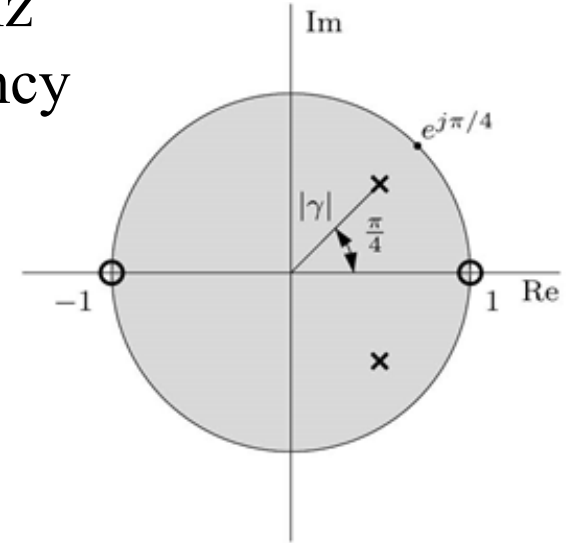
$$\Omega = \omega T = 2\pi f/f_s$$

$$p_1 = |\gamma|e^{j\pi/4} \quad p_2 = |\gamma|e^{-j\pi/4}$$

$$H(z) = K \frac{(z-1)(z+1)}{(z-|\gamma|e^{j\pi/4})(z-|\gamma|e^{-j\pi/4})} = K \frac{z^2-1}{z^2-\sqrt{2}|\gamma|z+|\gamma|^2}$$

$$H(e^{j\Omega}) = H(z)|_{z=e^{j\Omega}} = K \frac{e^{j2\Omega}-1}{e^{j2\Omega}-\sqrt{2}|\gamma|e^{j\Omega}+|\gamma|^2}$$

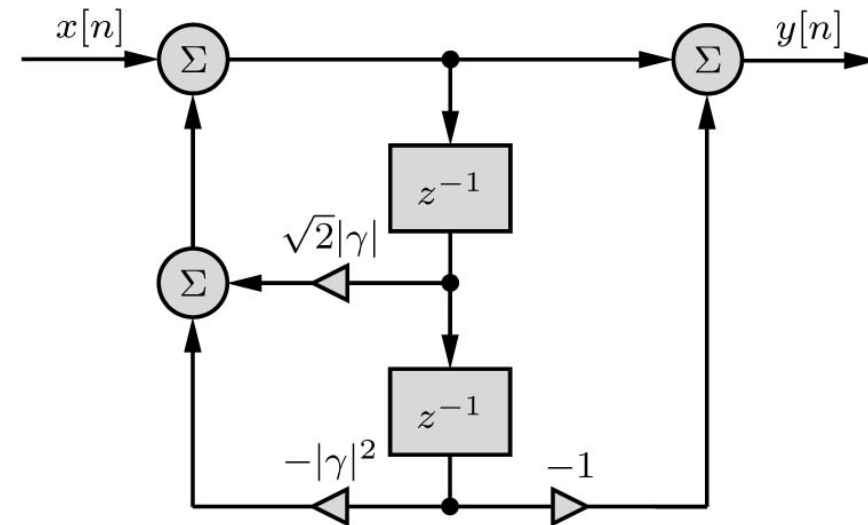
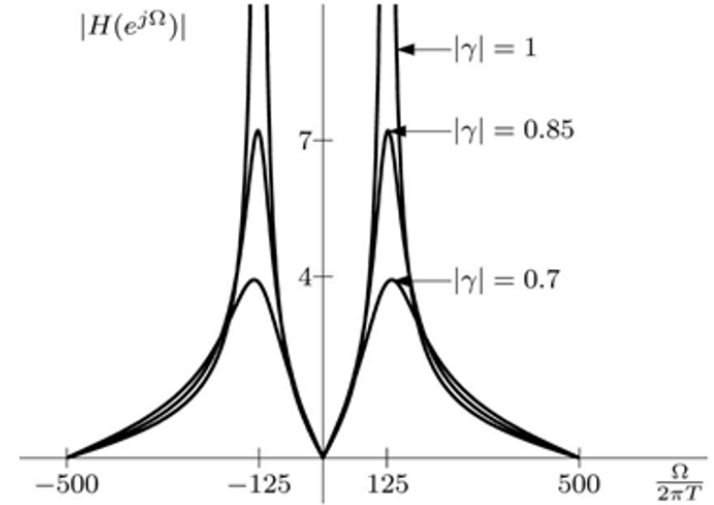
f	Ω
0	0
500	π
125	$\pi/4$



Continue Example

```
H = @(z,gamma) (z.^2-1)./(z.^2-sqrt(2)*abs(gamma)*z+(abs(gamma))^2);
T = 10^(-3);
Omega = linspace(-pi,pi,1001);
f = Omega/(2*pi*T);
z = exp(j*Omega);
plot(f,abs(H(z,0.7)),f,abs(H(z,0.85)),f,abs(H(z,1)));
axis([-500 500 -1 10]);
```

$$H(z) = K \frac{z^2 - 1}{z^2 - \sqrt{2}|\gamma|z + |\gamma|^2}$$



Example

Design a second-order notch filter with zero transmission at 250 Hz and a sharp recovery of gain to unity on both sides of 250 Hz. The highest significant frequency to be processed is $f_{\max} = 400$ Hz.

Solution

$$F_s \geq 2f_{\max} = 800 \text{ samples/sec}$$

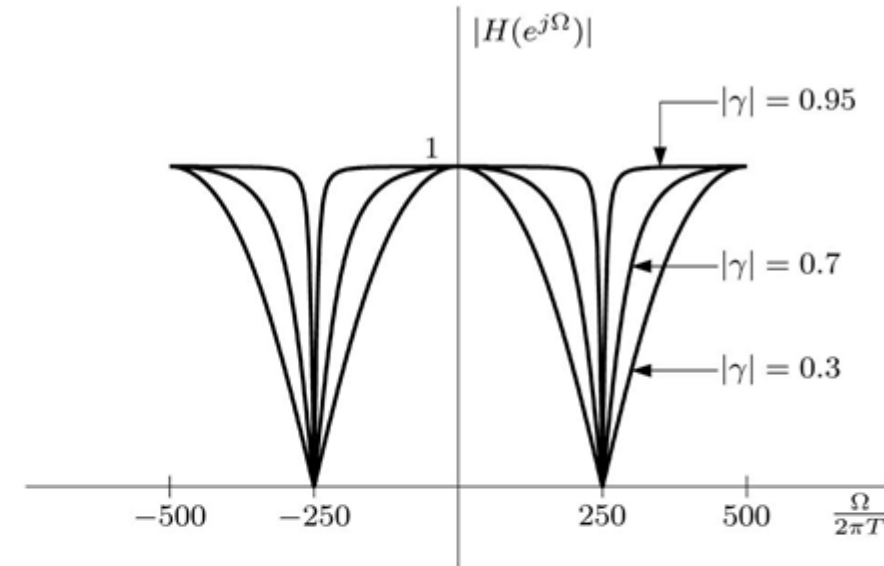
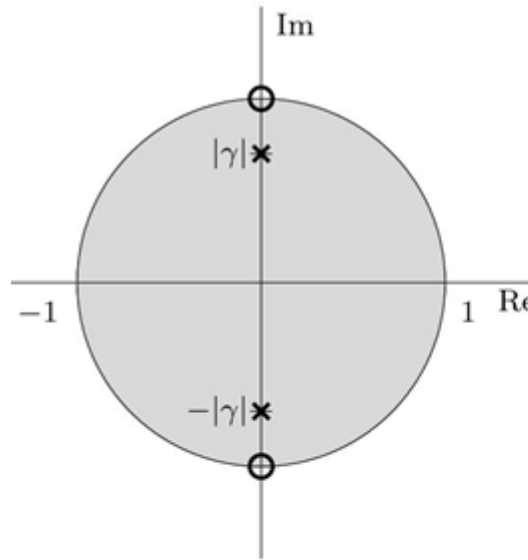
Choose $F_s = 1000$

f Ω

0 0

$\Omega = \omega / F_s = 2\pi f / F_s$ 400 0.8π

250 $\pi/2$



$$H(z) = K \frac{(z - j)(z + j)}{(z - j|\gamma|)(z + j|\gamma|)} = K \frac{z^2 + 1}{z^2 + |\gamma|^2}$$

```
H = @(z,gamma) (1+(abs(gamma)).^2)/2*(z.^2+1)./(z.^2+(abs(gamma))^2);
T = 10^(-3); Omega = linspace(-pi,pi,1001); f = Omega/(2*pi*T); z = exp(1j*Omega);
plot(f,abs(H(z,0.30)),f,abs(H(z,0.7)),f,abs(H(z,0.95))); axis([-500 500 -1 10]);
```

